SOFTWARE ENGINEERING ASSIGNMENT: SDLC MODEL SELECTION

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Course: Software Engineering

Topic: SDLC Model Selection

Objective: To analyse different software project scenarios and select the most suitable SDLC model with proper justification.

SDLC Model Selection for College Website

Introduction:

A College Website is an online platform that provides information about the institution such as courses offered, faculty details, admission process, notices, events, results, and contact information. It may also include features like student login, online forms, and announcements.

The system is mostly informational but may require periodic updates and minor feature enhancements.

Selected SDLC Model:

Waterfall Model: The Waterfall Model is a linear and sequential Software Development Life Cycle (SDLC) model in which development progresses through a series of well-defined phases. Each phase must be completed before moving on to the next, and there is no overlap between phases.

The main phases of the Waterfall Model include Requirement Analysis, System Design, Implementation, Testing, Deployment, and Maintenance. The output of one phase becomes the input for the next, forming a step-by-step flow similar to a waterfall.

Justification (Based on Given Factors):

1. Requirement Stability:

The requirements of a College Website are mostly stable and predefined, such as displaying course details, faculty information, admission process, and contact details. These do not change frequently, making the Waterfall model suitable.

2. Risk Level:

The project has a low risk level because it does not involve complex algorithms or real-time processing. Since the system is simple and predictable, Waterfall's linear approach works well.

3. Complexity:

A college website is less complex compared to systems like banking or e-commerce. It mainly involves static pages and basic functionalities, which can be easily developed in sequential phases.

4. User Involvement:

User involvement during development is minimal. Once requirements are gathered from the college administration, there is little need for continuous feedback, which aligns with the Waterfall model.

5. Flexibility: Flexibility requirements are low because major changes are not expected after development begins. Waterfall is suitable as it does not support frequent changes but works well with fixed requirements.

Other Supporting Factors:

Clear Documentation Process

The Waterfall model emphasizes detailed documentation at every stage. For a college website, proper documentation helps future developers easily maintain and update the system.

Easy Project Management

Since the phases are clearly defined (requirement → design → development → testing), it becomes easier to plan, monitor, and manage the project timeline.

Suitable for Academic Projects

Waterfall is widely used in academic environments because of its structured and easy-to-understand approach, making it ideal for a college website project.

Comparison with Other Models:

Agile Model:

- Not necessary because requirements are mostly fixed
- Frequent iterations are not required
- Can increase development time unnecessarily

Spiral Model:

- Too complex and expensive for a simple website
- Requires detailed risk analysis, which is not needed here

[Requirements]

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[System Design]

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[Implementation]

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[Testing]

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[Deployment]

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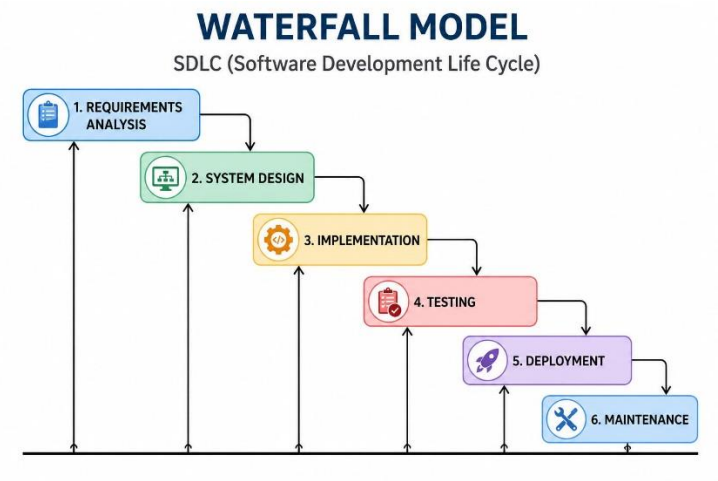
[Maintenance]

Conclusion:

The Waterfall model is the most suitable SDLC model for a College Website because it works well with stable and clearly defined requirements. Its structured approach ensures proper development, documentation, and maintenance of the system.